



Statistics of Financial Markets

Seminar 10: VaR, ES and backtesting



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Acronyms used in this material

- ▣ **AI:** Artificial Intelligence
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BVB:** Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ **CF:** Cornish–Fisher (expansion)
- ▣ **EODHD:** EOD Historical Data (market-data provider)
- ▣ **ES:** Expected Shortfall
- ▣ **EVT:** Extreme Value Theory
- ▣ **FHS:** Filtered Historical Simulation
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GPD:** Generalised Pareto Distribution
- ▣ **HS:** Historical Simulation
- ▣ **LR:** Likelihood Ratio (test)
- ▣ **OMV:** Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- ▣ **POF:** Proportion Of Failures (Kupiec test)
- ▣ **POT:** Peaks Over Threshold

Today's Question and Route

- **Question:** how large can tomorrow's loss be, and how do we check a risk number afterwards?
 - ▶ this seminar comes **before** Lecture 10: the section "Prerequisites for Today" gives every definition the tasks use
- Route
 - ▶ Part A: VaR and ES of a position, historical simulation, subadditivity, Kupiec and Christoffersen tests, on paper
 - ▶ Part B: five estimation methods, backtests per calendar year and two-asset portfolios on real data, each with an interpretation question
 - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class

Exercise Map

Task	Question	Type	Model
A1, A2	VaR 1% and ES 2.5% of a position: Normal and Student-t	Solved, Proposed	A1
A3, A4	historical simulation from the worst days of the BET; a subadditivity counterexample	Solved, Proposed	A3
A5, A6	the Kupiec test and the traffic light; the Christoffersen test	Solved, Proposed	A5
B1, B2	five estimation methods: BET; TLV, SNP, Bitcoin	Solved, Proposed	B1
B3, B4	backtests per calendar year: S&P 500; BET, Bitcoin	Solved, Proposed	B3
B5, B6	two-asset portfolios: TLV and SNP; DAX and S&P 500	Solved, Proposed	B5
C1, C2	is ES 2.5% as large as VaR 1% on real data? what is wrong in an AI answer?	Proposed	B1, A5

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data Used

Series	Source	Price used	Period
S&P 500, DAX, BET	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
Banca Transilvania (TLV), OMV Petrom (SNP), BRD	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; for portfolios, simple returns $R_t = 100(P_t/P_{t-1} - 1)$ on the common days; the last day is 18 September 2026
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, [Chapter 2](#))
- In the notebook: `returns('bet')`, `hs_var(x)`, `kupiec(x, n)` (package `arch` for GARCH); no account or key is needed

Prerequisites for Today (1/5): VaR and ES

- X : the daily return (P&L) of a position, in %; level α = the probability of the tail (1% or 2.5%)
 - ▶ q_α : the α -quantile of X , a negative number for small α
- **VaR** (value at risk): $\text{VaR}_\alpha = -q_\alpha$, the loss exceeded with probability α ; we write **VaR 1%**
 - ▶ a positive number: "VaR 1% = 3%" means a loss of more than 3% on one day in a hundred
- **ES** (expected shortfall): $\text{ES}_\alpha = -E[X \mid X \leq q_\alpha]$, the average loss on the worst α of days; we write **ES 2.5%**
 - ▶ $\text{ES}_\alpha \geq \text{VaR}_\alpha$ at the **same** level α
- **Historical simulation** (HS): sort the n returns, $r_{(1)} \leq \dots \leq r_{(n)}$, $k = \lceil n\alpha \rceil$ (rounded up)
 - ▶ $r_{(i)}$: the i -th smallest return; k : the number of days in the tail
 - ▶ $\widehat{\text{VaR}}_\alpha = -r_{(k)}$ (the k -th worst day), $\widehat{\text{ES}}_\alpha = -(r_{(1)} + \dots + r_{(k)})/k$ (the mean of the k worst days)
 - ▶ in money: $W \times \text{VaR}/100$ for a position of value W

Prerequisites for Today (2/5): Parametric VaR and ES

- **Normal** $N(\mu, \sigma^2)$: $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$, $\text{ES}_\alpha = -\mu + \sigma \varphi(z_\alpha)/\alpha$ (φ : the $N(0, 1)$ density)
 - ▶ μ, σ : mean and standard deviation; $z_\alpha = \Phi^{-1}(\alpha)$: the α -quantile of $N(0, 1)$; $z_{0.01} = -2.326$, $z_{0.025} = -1.960$, $\varphi(-1.960) = 0.0584$
 - ▶ h days (i.i.d., mean 0): $\text{VaR}^{(h)} = \sqrt{h} \text{VaR}^{(1)}$ (the square-root-of-time rule)
- **Standardised Student-t** with ν degrees of freedom (variance 1), $t_\alpha = t_\nu^{-1}(\alpha)$, f_ν the density of t_ν
 - ▶ $q_\alpha(z) = t_\alpha \sqrt{(\nu - 2)/\nu}$; $\text{ES}_\alpha(z) = \sqrt{(\nu - 2)/\nu} \frac{f_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$
 - ▶ with $X = \mu + \sigma z$: $\text{VaR}_\alpha = -(\mu + \sigma q_\alpha(z))$, $\text{ES}_\alpha = -\mu + \sigma \text{ES}_\alpha(z)$
- **Cornish–Fisher**: $z_{CF} = z + (z^2 - 1)S/6 + (z^3 - 3z)K/24 - (2z^3 - 5z)S^2/36$ (S : skewness, K : excess kurtosis)
 - ▶ in words: the Normal quantile $z = z_\alpha$ moved further into the left tail when $S < 0$ and $K > 0$
- **EVT** (POT, [Chapter 5](#)): $\text{VaR}_\alpha = u + \frac{\beta}{\xi} [(n\alpha/N_u)^{-\xi} - 1]$, GPD fitted to the losses above the threshold u
 - ▶ ξ, β : shape and scale of the GPD; n : number of days; N_u : number of losses above u

Prerequisites for Today (3/5): Coherence

- A risk measure ρ is **coherent** Artzner et al. (1999) if it is monotone, translation invariant, positively homogeneous and subadditive
 - ▶ L : a loss; $\rho(L)$: the capital needed to make it acceptable
 - ▶ **subadditivity**: $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$: diversification never adds risk
- ES is coherent Acerbi & Tasche (2002); VaR is not always subadditive
 - ▶ for a discrete loss L (default or not): VaR_α is the smallest l with $P(L > l) \leq \alpha$
 - ▶ ES_α = the average loss over the worst α of the probability mass (a part of an atom counts with its share)
- Two independent events with probabilities p : both happen with p^2 , exactly one with $2p(1 - p)$

Prerequisites for Today (4/5): Backtesting

- **Exception:** $r_t < -\text{VaR}_t$ (the VaR forecast made the day before); $I_t = 1$ on such days, else 0
 - ▶ correct VaR 1%: the I_t are i.i.d. Bernoulli(0.01); x exceptions in n days: $x \sim \text{Binomial}(n, 0.01)$
- **Kupiec** [Kupiec \(1995\)](#), $\hat{\pi} = x/n$: $LR_{uc} = -2[\ln L_0 - \ln L_1] \sim \chi^2(1)$
 - ▶ $\ln L_0 = (n - x) \ln 0.99 + x \ln 0.01$, $\ln L_1 = (n - x) \ln(1 - \hat{\pi}) + x \ln \hat{\pi}$
 - ▶ $\ln L_0, \ln L_1$: log-likelihoods of the exceptions with the rate 1% and with the observed rate $\hat{\pi}$; a large LR_{uc} : the rate is wrong
- **Christoffersen** [Christoffersen \(1998\)](#): n_{ij} = days with $I_{t-1} = i$, $I_t = j$; $\hat{\pi}_{01} = n_{01}/(n_{00} + n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10} + n_{11})$
 - ▶ $LR_{ind} = -2[\ln L(\hat{\pi}) - \ln L(\hat{\pi}_{01}, \hat{\pi}_{11})] \sim \chi^2(1)$; $LR_{cc} = LR_{uc} + LR_{ind} \sim \chi^2(2)$; critical values at 5%: 3.84 and 5.99
- **Basel traffic light** [Basel Committee \(1996b\)](#), 250 days of VaR 1%: green 0–4, yellow 5–9, red 10 or more exceptions
 - ▶ capital multiplier 3+ plus factor; plus factors for 5, 6, 7, 8, 9 exceptions: 0.40, 0.50, 0.65, 0.75, 0.85; red: 1

Prerequisites for Today (5/5): Portfolios and Dependence

- **Weights w_1, w_2 :** $R_p = w_1 R_1 + w_2 R_2$, $\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$
 - ▶ **variance–covariance VaR:** $-(\mu_p + z_\alpha \sigma_p)$; diversification benefit: $1 - \text{VaR}_p / (w_1 \text{VaR}_1 + w_2 \text{VaR}_2)$
- **Tail dependence:** $P(U_1 \leq q \mid U_2 \leq q)$ for small q , with U_i the ranks divided by $n + 1$ (pseudo-observations)
 - ▶ the probability that asset 1 has one of its worst days, given that asset 2 has one of its worst days
 - ▶ equals q under independence; correlation alone does not determine it
- **Copula:** the joint distribution of (U_1, U_2) ; Sklar: $F(x_1, x_2) = C(F_1(x_1), F_2(x_2))$
 - ▶ Gaussian copula: no tail dependence; t copula with ν : $\lambda = 2t_{\nu+1}(-\sqrt{(\nu+1)(1-\rho)/(1+\rho)}) > 0$ [Demarta & McNeil \(2005\)](#)
 - ▶ λ : the limit of the tail dependence as $q \rightarrow 0$; $t_{\nu+1}(\cdot)$: the Student-t distribution function; ρ from Kendall's τ : $\rho = \sin(\pi\tau/2)$

A1: VaR and ES of a Normal Position [Solved]

Task

- ▣ A position of 1 000 000 RON; daily returns $N(\mu, \sigma^2)$ with $\mu = 0.05\%$, $\sigma = 1.4\%$.
- ▣ 1. Compute VaR 1% in % and in RON.
- ▣ 2. Compute ES 2.5% in % and in RON.
- ▣ 3. Compute the 10-day VaR 1% with the square-root-of-time rule (μ ignored).
- ▣ 4. Compute the ratio ES 2.5% / VaR 1%.
- ▣ Report: six numbers and one sentence.

Solution

- ▣ 1. $-(0.05 - 2.326 \times 1.4) = -0.05 + 3.256 = 3.21\%$, i.e. 32 069 RON
- ▣ 2. $-0.05 + 1.4 \times 0.0584/0.025 = -0.05 + 3.270 = 3.22\%$, i.e. 32 229 RON
- ▣ 3. $\sqrt{10} \times 2.326 \times 1.4 = 10.30\%$, i.e. 102 992 RON
- ▣ 4. $3.22/3.21 = 1.005$
- ▣ For Normal returns ES 2.5% and VaR 1% are almost the same number.

A2: The Same Position with Student-t Returns [Proposed]

Solution discussed in class

Task

- ▣ As in A1, but the standardised returns are Student-t with $\nu = 4$: $X = \mu + \sigma z$; $t_4^{-1}(0.01) = -3.747$, $t_4^{-1}(0.025) = -2.776$, $f_4(-2.776) = 0.0256$. Model: A1.
- ▣ 1. Compute $q_{0.01}(z)$ and VaR 1% in % and in RON.
- ▣ 2. Compute $ES_{2.5\%}(z)$ and ES 2.5% in % and in RON.
- ▣ 3. Compare both numbers with A1.
- ▣ Report: six numbers and one sentence.

A3: Historical Simulation from the Worst Days of the BET [Solved]

Task

- ▣ The last 500 daily BET returns (12 September 2024 – 18 September 2026); the 15 smallest, in %: -3.68 ; -3.35 ; -3.12 ; -3.03 ; -2.93 ; -2.71 ; -2.61 ; -2.41 ;
- ▣ -2.38 ; -2.33 ; -2.27 ; -2.11 ; -2.08 ; -2.02 ; -1.93 .
- ▣ 1. Compute VaR 1% and VaR 2.5% by historical simulation.
- ▣ 2. Compute ES 2.5%.
- ▣ 3. Translate VaR 1% and ES 2.5% into RON for a position of 1 000 000 RON.
- ▣ Report: five numbers and one sentence.

Solution

- ▣ 1. $k = \lceil 500 \times 0.01 \rceil = 5$: VaR 1% = 2.93%;
 $k = \lceil 12.5 \rceil = 13$: VaR 2.5% = 2.08%
- ▣ 2. $ES\ 2.5\% = 35.01/13 = 2.69\%$ (the average of the 13 smallest)
- ▣ 3. VaR 1%: 29 300 RON; ES 2.5%: 26 931 RON
- ▣ Here ES 2.5% is below VaR 1%: ES is larger than VaR only at the same level; $ES\ 2.5\% \geq VaR\ 2.5\%$ holds.

A4: Two Bonds and Subadditivity [Proposed]

Solution discussed in class

Task

- Two independent bonds; each loses 100 with probability 3% and 0 otherwise; level 5%. Model: A3 and slide 3/5.
- 1. Compute VaR 5% and ES 5% of one bond.
- 2. Write the distribution of the total loss of the two bonds.
- 3. Compute VaR 5% and ES 5% of the total loss.
- 4. Check subadditivity for VaR and for ES.
- Report: four numbers and two sentences.

A5: The Kupiec Test and the Traffic Light [Solved]

Task

- ▣ A bank's VaR 1% was exceeded on 7 of the last 250 days. A second model was exceeded on 16 of 1000 days.
- ▣ 1. Compute $\ln L_0$, $\ln L_1$ and LR_{uc} for the bank, and decide at 5%.
- ▣ 2. Give the Basel zone, the plus factor and the capital multiplier.
- ▣ 3. Compute LR_{uc} for the second model and decide at 5%.
- ▣ Report: five numbers, two decisions and one sentence.

Solution

- ▣ 1. $243 \ln 0.99 + 7 \ln 0.01 = -34.68$;
 $243 \ln 0.972 + 7 \ln 0.028 = -31.93$; $LR_{uc} = 5.50 > 3.84$ ($p = 0.019$): rejected
- ▣ 2. Yellow zone ($P(X \leq 6) = 98.63\%$,
 $P(X \leq 7) = 99.60\%$); plus 0.65, multiplier 3.65
- ▣ 3. $984 \ln 0.99 + 16 \ln 0.01 = -83.57$;
 $984 \ln 0.984 + 16 \ln 0.016 = -82.03$; $LR_{uc} = 3.08$ ($p = 0.079$): not rejected
- ▣ A rate of 1.6% is not rejected with 1000 days; 2.8% is rejected with 250: the test needs a large deviation or many days.

A6: The Christoffersen Test [Proposed]

Solution discussed in class

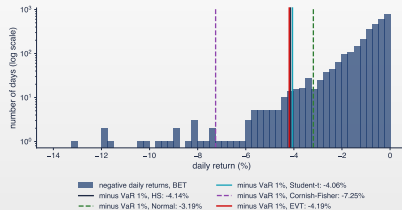
Task

- ▣ 1000 days of VaR 1% forecasts give the transition counts $n_{00} = 978$, $n_{01} = 8$, $n_{10} = 8$, $n_{11} = 5$. Model: A5.
- ▣ 1. Compute $\hat{\pi}_{01}$, $\hat{\pi}_{11}$ and $\hat{\pi}$.
- ▣ 2. Compute LR_{ind} and decide at 5%.
- ▣ 3. Compute LR_{uc} (with $x = 13$, $n = 999$) and LR_{cc} , and decide at 5%.
- ▣ Report: six numbers, three decisions and one sentence.

B1: Five Methods for the BET [Solved]

- ▣ **Question:** how much do VaR 1% and ES 2.5% of the BET depend on the estimation method?
- ▣ **Inputs:** BET closes since 2000, daily log returns in %
- ▣ **Tasks**
 1. Compute VaR 1% and ES 2.5% by historical simulation, Normal and Student-t (maximum likelihood).
 2. Compute the skewness and the excess kurtosis, and the Cornish–Fisher VaR 1%.
 3. Fit a GPD to the losses above their 90% quantile and compute the EVT VaR 1% and ES 2.5%.
 4. Draw the left tail with the five values of minus VaR 1%.
 5. Interpretation: which number would you report to a risk committee?
- ▣ **Report:** a table of ten numbers, the chart and two sentences
- ▣ **Notebook:** section B1

B1: Solution [Solved]



	HS	Normal	Student-t	Cornish-Fisher	EVT
VaR 1%	4.14	3.19	4.06	7.25	4.19
ES 2.5%	4.57	3.21	4.76	7.93	4.57

- $n = 6\,670$; $\hat{\mu} = 0.064$, $\hat{\sigma} = 1.40$; $S = -0.65$, $K = 11.1$; $\hat{\nu} = 2.43$; EVT: $u = 1.24\%$, $N_u = 667$, $\hat{\xi} = 0.23$
- The Normal VaR 1% is 23% below the data; Cornish-Fisher ($z_{CF} = -5.23$) is far above all others
- Interpretation: report the historical or the EVT values, which agree (VaR 1% about 4.14%, ES 2.5% about 4.57%); mention that the Normal understates and Cornish-Fisher is outside its range with $K > 10$

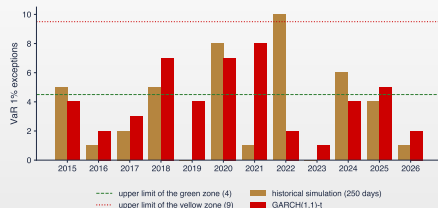
B2: TLV, SNP and Bitcoin [Proposed]

- **Question:** does the ranking of the methods found for the BET hold for two BVB stocks and for Bitcoin? Model: B1.
- **Inputs:** TLV and SNP since 2010 (TLV without 30–31 May 2016), Bitcoin since 2014; daily log returns in %
- **Tasks**
 1. Compute VaR 1% and ES 2.5% of each series by the five methods of B1.
 2. Compute by how much the Normal VaR 1% is below the historical VaR 1%.
 3. Report $\hat{\nu}$, $\hat{\xi}$ and the excess kurtosis of each series.
 4. Interpretation: for which series does the Normal understate VaR 1% most?
- **Report:** one table and two sentences
- **Notebook:** section B2

B3: Backtesting the S&P 500 Year by Year [Solved]

- ▣ **Question:** which of two VaR 1% models would a supervisor have penalised, year by year, since 2015?
- ▣ **Inputs:** S&P 500 since 2000; forecasts for every day since 2 January 2015
- ▣ **Tasks**
 1. Compute the one-day VaR 1% by historical simulation on the last 250 days.
 2. Compute the one-day GARCH(1,1)-t VaR 1%, re-estimating the model at the start of every year on all earlier data.
 3. Count the exceptions of each model in every calendar year and give the Basel zone of each year.
 4. Run the Kupiec and Christoffersen tests on the whole period.
 5. Interpretation: which model would a supervisor have penalised in 2022?
- ▣ **Report:** the chart, six statistics with p-values and two sentences
- ▣ **Notebook:** section B3

B3: Solution [Solved]



	n	x	rate (%)	LR_{uc} (p)	LR_{ind} (p)	LR_{cc} (p)	red/yellow years
HS 250	2 944	43	1.46	5.52 (0.019)	23.04 (< 0.001)	28.56 (< 0.001)	1/4
GARCH-t	2 944	49	1.66	10.94 (< 0.001)	6.78 (0.009)	17.72 (< 0.001)	0/4

- Both models are exceeded too often (expected about 29.4); HS exceptions cluster strongly ($n_{11} = 7$, p ind. < 0.001), GARCH-t less ($n_{11} = 4$)
- Interpretation: in 2022 HS had 10 exceptions (red zone), GARCH-t 2: the slow window was penalised when volatility rose; GARCH-t paid in calmer years (2021: 8)

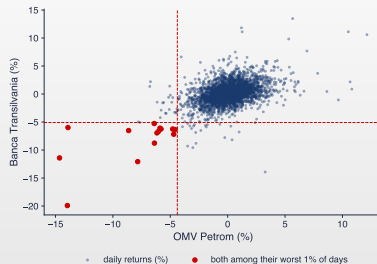
B4: Backtesting the BET and Bitcoin [Proposed]

- **Question:** does the conclusion of B3 hold on the BVB and for Bitcoin? Model: B3.
- **Inputs:** BET since 2000, forecasts since 2015; Bitcoin since 2014, forecasts since 2016
- **Tasks**
 1. Compute both VaR 1% forecasts as in B3.
 2. Count the exceptions per calendar year and give the zones.
 3. Run the Kupiec and Christoffersen tests on the whole period.
 4. Interpretation: which model would you keep for each series?
- **Report:** one table and two sentences
- **Notebook:** section B4

B5: A Portfolio of Two BVB Stocks [Solved]

- **Question:** how much does diversification between Banca Transilvania and OMV Petrom reduce VaR 1%, and does it survive a crisis?
- **Inputs:** TLV and SNP adjusted closes since 2010, joined on common days; simple returns in %; weights 50/50
- **Tasks**
 1. Compute the variance–covariance VaR 1% of the portfolio, the two individual VaRs and the diversification benefit.
 2. Compute the historical VaR 1% and ES 2.5% of the portfolio.
 3. Compute the correlation in 2017 and in February–May 2020, and the tail dependence at $q = 5\%$ and $q = 1\%$.
 4. Fit a t copula and compute the copula VaR 1% with empirical margins.
 5. Interpretation: why does the variance–covariance VaR understate the risk of this portfolio?
- **Report:** eight numbers, the chart and two sentences
- **Notebook:** section B5

B5: Solution [Solved]



- ▣ $n = 3\,461$; $\hat{\sigma}$: TLV 1.84, SNP 1.70; $\hat{\rho} = 0.46$; $\sigma_p = 1.511$; VaR 1%: TLV 4.17, SNP 3.85, portfolio 3.41%; benefit 15%
- ▣ Historical: VaR 1% = 4.12%, ES 2.5% = 4.51%; t copula ($\hat{\nu} = 5.00$, $\lambda = 0.166$): VaR 1% = 3.96%
- ▣ Correlation: 2017 0.32, spring 2020 0.69; tail dependence 0.36 ($q = 5\%$), 0.40 ($q = 1\%$); joint worst-1% days: 14, against 0.35 under independence
- ▣ Interpretation: the Normal VaR is 17% below the historical one: both margins have heavy tails and the two stocks fall together on the worst days, when the correlation is highest

B6: DAX and S&P 500 [Proposed]

- **Question:** is the Gaussian copula good enough for a 50/50 DAX and S&P 500 portfolio? Model: B5.
- **Inputs:** DAX and S&P 500 closes since 2000, joined on common days; simple returns in %
- **Tasks**
 1. Compute the variance–covariance and the historical VaR 1% and ES 2.5%.
 2. Fit the Gaussian and the t copula and compare them with the LR statistic.
 3. Compute the copula VaR 1% and ES 2.5% with empirical margins for both copulas.
 4. Interpretation: is the Gaussian copula good enough for this pair?
- **Report:** one table and two sentences
- **Notebook:** section B6

C1: Is ES 2.5% as Large as VaR 1%? [Proposed]

- ▣ **Question:** Basel replaced VaR 1% by ES 2.5% because the two are equal for Normal returns; how close are they on real data and in crises?
- ▣ **Inputs:** S&P 500, DAX, BET, Bitcoin, TLV, SNP, BRD; models: B1, A5
- ▣ **Tasks**
 1. Compute the ratio ES 2.5% / VaR 1% by historical simulation on the whole sample of each series.
 2. Compute the same ratio on the 500 days that end in March 2009 and in May 2020.
 3. Compute the ratio implied by a fitted Student-t distribution.
 4. Interpretation: is ES 2.5% a stricter rule than VaR 1% on these markets?
- ▣ **Report:** a table and a plan for a project
- ▣ **Notebook:** section C1

C2: Audit an AI Answer [Proposed]

- ▣ A student asked an AI assistant to summarise the risk of the BET over the last 500 days. The answer:
- ▣ (a) The VaR 99% of the BET is -2.93% .
- ▣ (b) ES is always at least as large as VaR, so ES 2.5% is above 2.93%.
- ▣ (c) The 10-day VaR 1% is $10 \times 2.93 = 29.3\%$.
- ▣ (d) With 8 exceptions in 250 days the Kupiec p-value is 0.005, but the model stays in the green zone.
- ▣ (e) VaR is subadditive, so the VaR of a portfolio never exceeds the sum of the VaRs of its parts.
- ▣ (f) If the exceptions cluster, the Christoffersen test can reject even when their number is right.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

What You Should Take from Today

- ▣ $\text{VaR } 1\% = -q_{1\%}$, a positive loss; $\text{ES } 2.5\%$ averages the worst 2.5% of days; $\text{ES} \geq \text{VaR}$ only at the same level
- ▣ On real data the Normal method understates VaR; Cornish–Fisher fails for large kurtosis; HS and EVT agree at 1%
- ▣ Backtesting needs two checks: the number of exceptions (Kupiec) and their independence (Christoffersen)
- ▣ Diversification is weakest in crises: correlation and tail dependence rise together
- ▣ An AI answer is a draft: check the sign and the level of VaR, the horizon rule and the Basel zone

After the Seminar

- Lecture 10 develops each topic of today: definitions and coherence, estimation methods, conditional VaR, portfolios and copulas, backtesting
- Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- C1 can grow into a team project: rolling ratios, bootstrap intervals, the capital under each rule
- Reading: [Franke, Härdle & Hafner \(2019\)](#), Ch. 16; exercises in [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 16; [McNeil, Frey & Embrechts \(2015\)](#), Ch. 2

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